


CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

Test Pattern

JEE(Advanced)
FULL SYLLABUS
10-04-2024
JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY
GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 15)

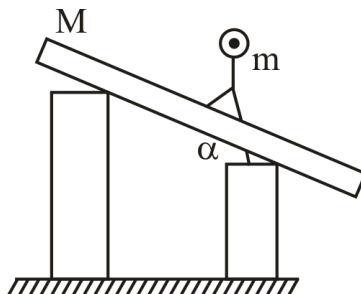
- This section contains **FIVE (05)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

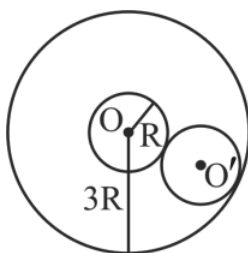
Negative Marks : -1 In all other cases

1. A plank is held at an angle α to the horizontal on two fixed supports A and B. The plank can slide against the supports (with out friction) because of its weight Mg . With what acceleration and in which direction, a man of mass m should move so that the plank does not move?

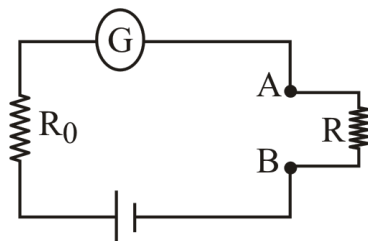


- (A) $g \sin \alpha \left(1 + \frac{m}{M}\right)$ down the incline
- (B) $g \sin \alpha \left(1 + \frac{M}{m}\right)$ down the incline
- (C) $g \sin \alpha \left(1 + \frac{m}{M}\right)$ up the incline
- (D) $g \sin \alpha \left(1 + \frac{M}{m}\right)$ up the incline
2. With standard resistances 2.0Ω and 1.0Ω in the gaps of a metre bridge, the balancing length is 66.9 cm from the higher resistance A of the bridge wire. When the resistances are interchanged, the balancing length is 33.5 cm from the same end A. With an unknown resistance X and standard 1.0Ω in the gaps, the balancing length is 59.7 cm from the end A nearer to X . Then value of X is
- (A) 0.675Ω (B) 0.670Ω
- (C) 0.680Ω (D) 1.481Ω

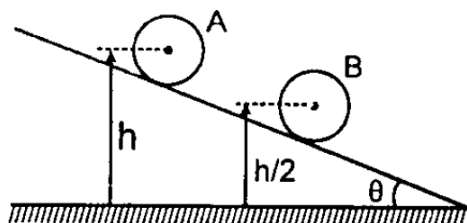
3. A thick hollow sphere with inner radius R and outer radius $3R$ has a uniform volume density ρ . It has a spherical cavity of radius R as shown. The gravitational field at the centre O' of the cavity is



- (A) zero
(B) $\frac{2}{3}\pi G\rho R$
(C) $\frac{5}{3}\pi G\rho R$
(D) $\frac{7}{3}\pi G\rho R$
4. Circuit shown in figure is a simple ohm – meter, in which G is a galvanometer, R_0 is known resistance and R is the resistance which is to be measured. If A and B are short circuited by a resistanceless wire, galvanometer gives full scale deflection. Then to read the resistance R directly from galvanometer, ohm – meter scale would look like:



- (A)
- (B)
- (C)
- (D)
5. Two identical balls A & B of mass m each are placed on a fixed wedge as shown in figure. Ball B is kept at rest and it is released just before two balls collide. Ball A rolls down without slipping on inclined plane & collide elastically with ball B . The kinetic energy of ball A just after the collision with ball B is :



- (A) $\frac{mgh}{7}$
(B) $\frac{mgh}{2}$
(C) $\frac{2mgh}{5}$
(D) $\frac{7mgh}{5}$

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

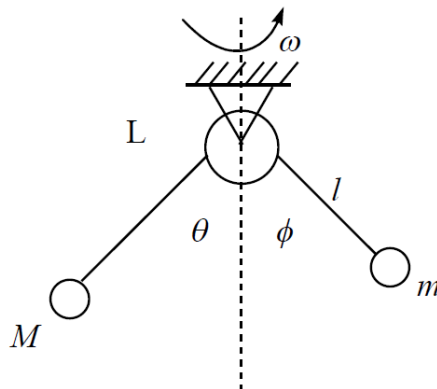
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

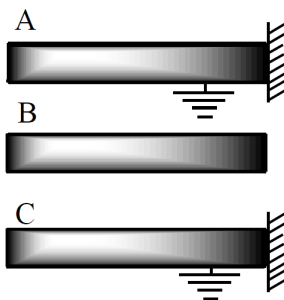
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6. An open bottle lies on a precise weighing scale. A fly then enters the bottle and flutters around in futile attempt to escape. Another fly, instead of using its wings to fly, uses a small helium filled balloon fitted to it by a microartist, in order to remain airborne. Choose the correct option(s) from the following statements:
- (A) reading will more in first case when it flutters
- (B) reading will more in second case when it is using balloon
- (C) reading in second cases is higher than reading before the fly entered the bottle.
- (D) reading will be same in both cases and equal to reading before any fly entered the bottle

7. A double conical pendulum consists of two masses, m and M , connected by a massless string passing over a frictionless, massless pulley. The entire apparatus rotates clockwise and freely at constant angular speed ω (rad/s) about the vertical axis (dashed line) passing through centre of pulley as shown. After the system comes in steady state, the length of string on either sides of pulley are l and L . Pick up the correct option from the following:

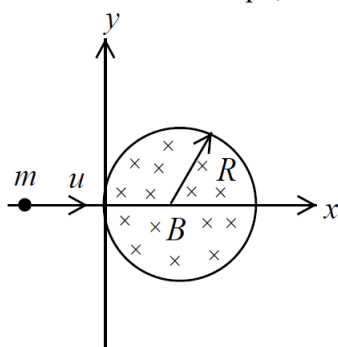


- (A) $\frac{\cos \phi}{\cos \theta} = \frac{m}{M}$
- (B) $\cos \theta = \frac{g}{\omega^2 L}$
- (C) $Ml = mL$
- (D) $\cos \phi = \frac{g}{\omega^2 l}$
8. In case of superposition of waves (at $x = 0$)
- $y_1 = 2\sin(1026\pi t)$
- $y_2 = 2\sin(1014\pi t)$
- (A) the frequency of resulting wave is 510 Hz
- (B) the amplitude of resulting wave varies at frequency 3 Hz
- (C) the frequency of beats is 6 Hz
- (D) the amplitude of resulting wave varies at frequency 6 Hz

9. A, B and C are three large, parallel conducting plates, placed horizontally. A and C are rigidly fixed and earthed. B is given some charge. Under electrostatic and gravitational forces, B may be:



- (A) in equilibrium midway between A and C
 (B) in equilibrium if it is closer to A than to C
 (C) in equilibrium if it is closer to C than to A
 (D) B can never be in stable equilibrium
10. A particle of specific charge $q/m = -2 \times 10^{10} \text{ C/kg}$ is travelling with a uniform velocity $v_0 \hat{i}$. A uniform steady magnetic field $\vec{B}$ is suitably applied for a duration of time $\Delta t = \pi \mu \text{ s}$ so that the particle's velocity just after removal of the field becomes $-v_0 \hat{i}$. If the motion of the particle is confined to the XOY plane, $\vec{B}$ is possibly
- (A) $50 \hat{k} \mu \text{ T}$ (B) $-100 \hat{k} \mu \text{ T}$
 (C) $+150 \hat{k} \mu \text{ T}$ (D) $-250 \hat{k} \mu \text{ T}$
11. A particle of charge 'q' and mass 'm' moving with speed 'u' along x-axis at origin enters the circular region of a uniform magnetic field $\vec{B} = B_0 \hat{k}$. The particle enters the region from origin at $t = 0$ and comes out of the region from point having co-ordinates (x, y, 0) at time $t = t_0$ with speed 'v'. If the region is gravity free space and the radius of circular region is $R = \frac{mu}{qB_0} \sqrt{3}$ then



- (A) $x = \frac{mu}{qB_0} \left(\frac{\sqrt{2}-1}{\sqrt{2}} \right), y = \frac{mu}{\sqrt{2}(qB_0)}$
 (B) $v = u$
 (C) $t_0 = \frac{\pi m}{2qB_0}$
 (D) $t_0 = \frac{2\pi m}{3qB_0}$

12. A conductor is made of an isotropic material (resistivity ρ) has rectangular cross-section. Horizontal dimension of the rectangle decreases linearly from $2x$ at one end to x at the other end and vertical dimension increases from y to $2y$ as shown in figure. Length of the conductor along the axis is equal to l . A battery is connected across the length of this conductor. Then:
- (A) Resistance of the conductor is equal to $\frac{2\rho l}{xy} \log_e(2)$
- (B) Rate of generation of heat per unit length is maximum at middle cross - section.
- (C) Drift velocity of conduction electrons is minimum at middle section.
- (D) At the ends of the conductor, electric field intensity is same.
13. A dipole of dipole moment p is placed at a distance $2R$ from the centre of a neutral conducting sphere of radius R . The direction of dipole is towards the centre of the sphere. A tangent is drawn from the dipole to the sphere which meets the sphere at point A.
- (A) The potential at point A is $p/16\pi\epsilon_0 R^2$
- (B) The potential at point A is $\sqrt{3}p/24\pi\epsilon_0 R^2$
- (C) The potential at point A due to induced charges is $(3 - 2\sqrt{3}) p/48\pi\epsilon_0 R^2$
- (D) The potential at point A due to dipole is $\sqrt{3}p/24\pi\epsilon_0 R^2$

Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

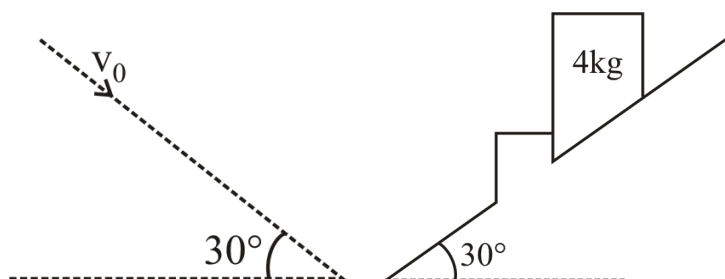
SECTION-III : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

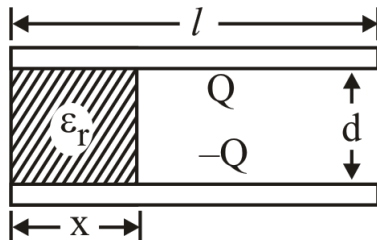
Full Marks : +3 If ONLY the correct answer is chosen.

Zero Marks : 0 In all other cases.

- Two particles A and B can execute SHM of amplitude 'a' each and time periods 4s and 6s respectively along the x-axis with origin as their equilibrium position. At time $t=0$, the particle A is at $x = -a$, B is at $x = 0$ and both are moving in the same direction. The instant of time at which they collide with each other for the first time is $9/n$ sec. find n
- A source of sonic oscillations with frequency $f_0 = 1700$ Hz and a receiver are located at the same point. At the moment $t = 0$ the source starts receding from the receiver with constant acceleration $a = 10.0 \text{ m/s}^2$. Assuming the velocity of sound to be equal to $v = 340 \text{ m/s}$. The oscillation frequency registered by the stationary receiver at $t = 10.0$ second after the start of motion is $50n$ Hz less than the natural frequency of the source. find n.
- A 20 gm bullet is fired into a 4 kg wooden block and becomes embedded in it. Knowing that the block and bullet then move up the smooth long inclined (fixed) for 12 mili sec before they come to a stop. The magnitude of the initial velocity of the bullet is $4n \text{ m/sec}$, find n (take $g = 10 \text{ m/s}^2$)



4. Two conducting square plates with length l are arranged parallel to each other at a distance d ($\ll l$). The space between the plates is filled with a dielectric of relative permittivity $\epsilon_r = 4$ up to a length $x = l/3$. The upper plate is given a charge Q and the lower plate is given a charge $-Q$ as shown in the figure. If $Q = 6\mu\text{C}$ then find how much bound charge (in μC) is present on the lower surface of dielectric?



5. A container filled with air under pressure P_0 contains a soap bubble of radius R . The air pressure has been reduced to half isothermally and the new radius of the bubble becomes $\frac{5R}{4}$. If the surface tension of the soap water solution is S , P_0 is found to be $\frac{16nS}{R}$ SI unit. Find n .

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 15)

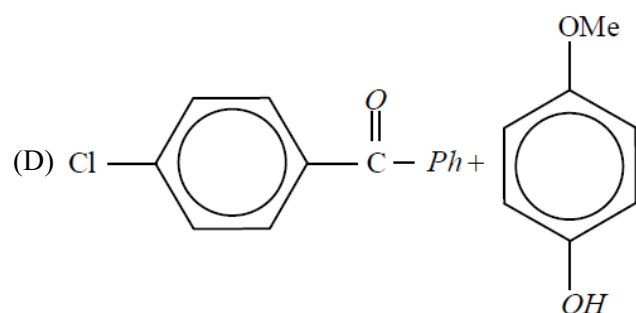
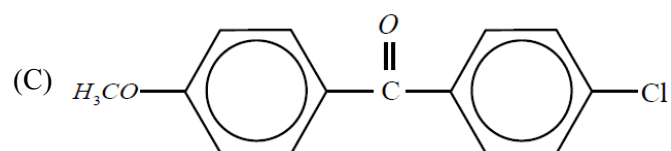
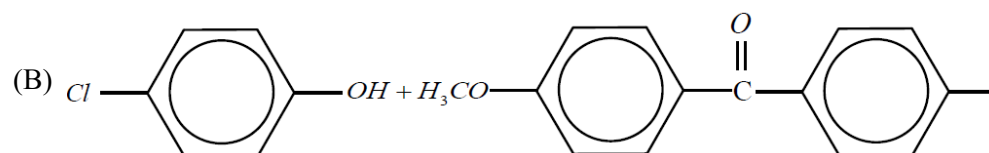
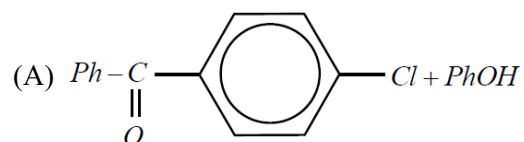
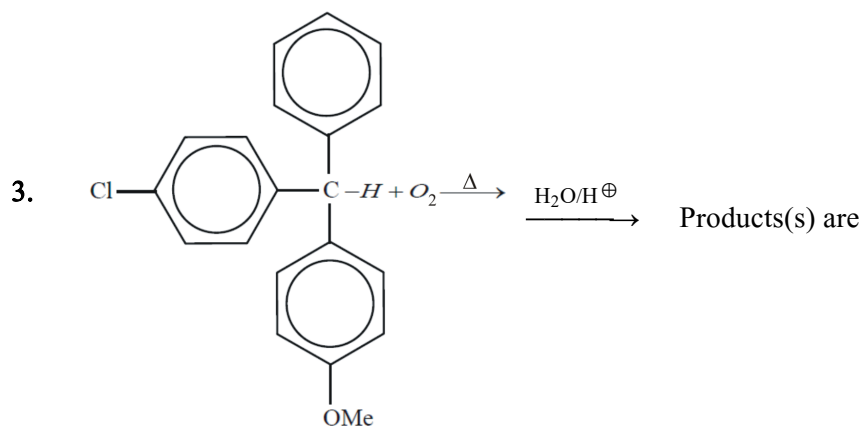
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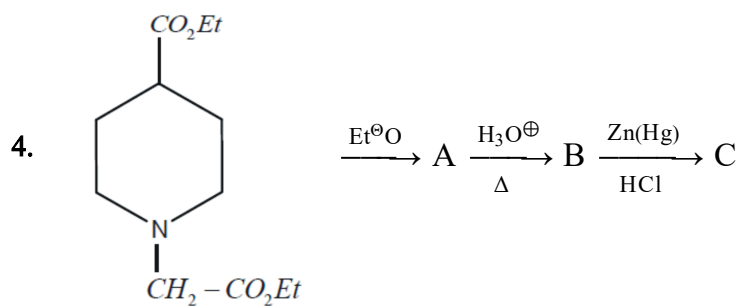
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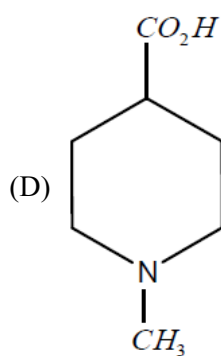
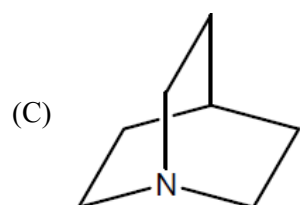
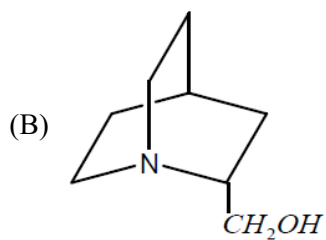
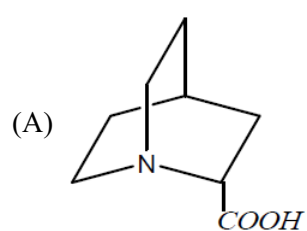
Negative Marks : -1 In all other cases

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1. The correct statement among the following is
 - (A) When sodium nitrate is heated with sodium metal the products formed are Na_2O and N_2O
 - (B) When sodium nitrite is heated with sodium metal the products formed are Na_2O and N_2
 - (C) When sodium nitrate is fused with sodium metal the products formed are NaNO_2 and Na_2O
 - (D) When sodium nitrite is heated with sodium metal the products formed are Na_2O and NO
 2. A deliquescent white crystalline hydroxide X react with a metal nitrate (Y) to form another hydroxide which decomposes to give insoluble brown precipitate. (X) and (Y) are
 - (A) NaOH , $\text{Zn}(\text{NO}_3)_2$
 - (B) NaOH , AgNO_3
 - (C) $\text{Ca}(\text{OH})_2$, $\text{Hg}(\text{NO}_3)_2$
 - (D) NaOH , $\text{Al}(\text{NO}_3)_3$





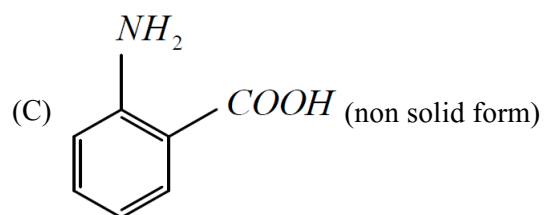
C is:



5. Which of the following does not exist as a Zwitter ion?

(A) Glycine

(B) Para Sulphanilic acid



(D) Alanine

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
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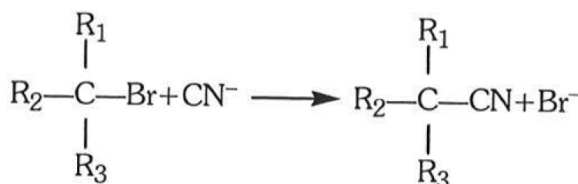
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

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6. When KF is dissolved in HF, a new species KHF_2 is formed. The type of bond present in KHF_2 is/are
- (A) Ionic bond
(B) Covalent bond
(C) Coordinate covalent bond
(D) Hydrogen bond
7. $\text{HNO}_3 + \text{P}_4\text{O}_{10} \rightarrow \text{A} + \text{HPO}_3$
 $\text{A} + \text{Na} \rightarrow \text{B} + \text{C}$
 $\text{Cu} + \text{HNO}_3 (\text{Conc}) \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{C} + \text{H}_2\text{O}$
 $\text{C} + \text{NaOH} \rightarrow \text{B} + \text{D}$
Select the correct statements
- (A) B acts as a base in molten A
(B) D acts as an acid in molten A
(C) C has very less tendency toward dimerisation at room temperature
(D) All the atoms in A are in same plane

8. Which of the following ions do not contain electron in ns (4s) orbital
- (A) Mn^+ (B) Cr^+
(C) V^+ (D) Fe^+
9. Which of the following statements are correct
- (A) The transition metal ion is more polarizing compared to S-block metal ions of comparable size for a particular oxidation state
- (B) The solubility of silver halides is in the order $\text{AgF} > \text{AgCl} > \text{AgBr} > \text{AgI}$ and the decrease in solubility is due to increased covalent character by the increased polarisability of halide ions
- (C) LiCl is soluble in organic solvents while NaCl is insoluble in such solvents due to less covalent character
- (D) Thermal stability of oxosalts decreases with decrease in the size of cation
10. The energy profiles for the two possible mechanisms of the following reaction



Are given below

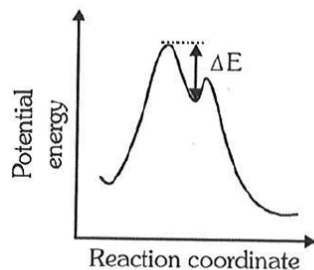


Figure X

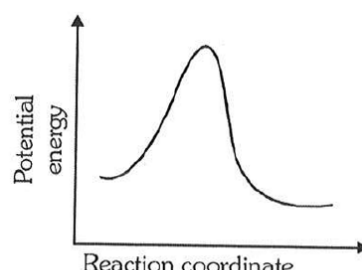


Figure Y

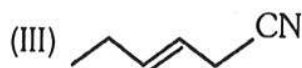
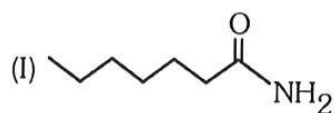
Consider the following substrates (I to IV) for the above reaction

- (I) $\text{R}_1 = \text{R}_2 = \text{R}_3 = \text{H}$
(II) $\text{R}_1 = \text{C}_2\text{H}_5$; $\text{R}_2 = \text{R}_3 = \text{H}$
(III) $\text{R}_1 = \text{C}_2\text{H}_5$; $\text{R}_2 = \text{CH}_3$; $\text{R}_3 = \text{H}$
(IV) $\text{R}_1 = \text{C}_2\text{H}_5$; $\text{R}_2 = \text{R}_3 = \text{CH}_3$

Choose the correct statements

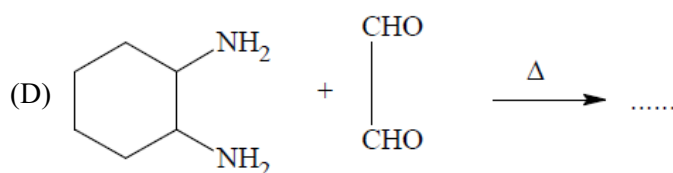
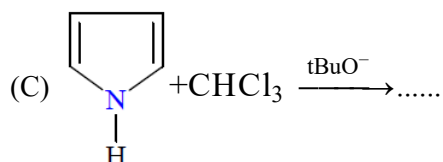
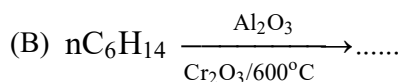
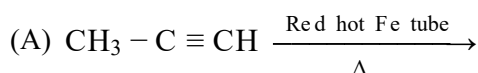
- (A) The reaction mechanism changes from the one shown in figure Y to that in figure X, as one changes the substrate from I to IV
- (B) Considering the mechanism corresponding to figure X for substrates I to IV, the correct order of ΔE is $\Delta E_{\text{I}} < \Delta E_{\text{II}} < \Delta E_{\text{III}} < \Delta E_{\text{IV}}$
- (C) The reaction for substrate I follows 2nd order kinetics and that for IV follows 1st order kinetics
- (D) If CN^- is taken in large excess, it is difficult to ascertain if the reaction follows mechanism corresponding to energy profile in figure X or that in figure Y.

11. The nitration of toluene takes place at the ortho, para and meta position, giving ortho-, para-, and meta nitrotoluenes, respectively. Assume that the meta product is formed in negligible amount and that the ortho and para substitutions are irreversible first – order reactions. Let the initial toluene concentration be $[A]_0$ and k_1 and k_2 be the ortho- and para – rate constants, respectively. Choose the correct statements
- (A) The rate of formation of the ortho substituted product is given by $k_1 [A] e^{-(k_1+k_2)t}$, where t is the time.
- (B) If the rate constant for the ortho product formation is twice that for the para product formation, then $[O\text{-nitrotoluene}] = 2 [p\text{-nitrotoluene}]$, at any time
- (C) Nitration at the ortho – and para – positions is favorable compared to that at the meta position as the ortho and para – intermediates are more stabilized than the meta – intermediate
- (D) If the methyl group of toluene is replaced by tertiary butyl group, more ortho product is expected
12. The following four compounds are subjected to different reactions



Choose the correct statements for these reactions

- (A) Compound I on reaction with bromine and NaOH gives the same compound as that obtained by complete reduction of III.
- (B) The compound that can be reduced by hydrogen under pressure in the presence of Raney Ni is IV.
- (C) Compound III on reaction with LiAlH_4 gives a compound which reacts with bromine under suitable condition.
- (D) The compound obtained by the reaction of LiAlH_4 with II is same as that obtained on heating IV with dilute H_2SO_4 at high temperature.
13. Which of the following will produce aromatic compound as product?



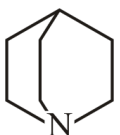
Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct answer is chosen.

Zero Marks : 0 In all other cases.

1. Chlorine oxidises hypo to a product (X). The difference in the sum of the oxidation numbers of sulphur in hypo and sulphur in product (X) is
2. A sulphate of a metal (M) on heating evolves two gases (B) and (C) and an oxide (D). Gas (B), turns acidified dichromate paper green while gas (C) forms a trimer.
Compound (D) forms a Lewis acid (E) with HCl (g). If the number of S – O – S bonds in trimer of gas (C) be 'n' and the number of equivalent M – Cl bonds in one molecule of (E) in vapour phase be 'm' then (m+n) is
3. Total number of possible isomers of the complex $[\text{Cr}(\text{en})_3][\text{Cr}(\text{OX})_3]$ is
4. How many of the following is/are condensation polymer? Teflon, Nylon 66, PVC, Buna-S, Bakelite, Terylene, Neoprene, Silicones, HDPE.
5. How many are the correct statements ?
 - 1) $\text{CH}(\text{CN})_3$ is more acidic than Picric acid
 - 2) Tri ethyl amine is more basic than  (Quinuclidine)
 - 3) In propene the eclipsed conformation lies lesser potential energy than bisected Conformation
 - 4) NH_4Cl , naphthalene, I_2 , indigo are all compounds which sublime easily

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. An ellipse in the first quadrant is tangent to co-ordinate axes. If one of focus is $F_1(3,7)$, and the other focus is $F_2(d,7)$, then the value of $\sqrt{3d}$ is equal to
 (A) 5 (B) 7
 (C) 14 (D) 49
2. Let $f : \mathbb{R} \rightarrow \mathbb{R}^+$ be a continuous function such that $f(x) = 2(1+x^2) \left(1 + \int_0^x \frac{f(t)}{1+t^2} dt\right)$, $\forall x \in \mathbb{R}$.
 Then
 (A) $f(2)=6e^4$ (B) $f(1)=4e$
 (C) $f(2)=10e^4$ (D) $f(3)=8e^3$
3. $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \sum_{r=1}^n \frac{1}{\sqrt{r} + \sqrt{r^2 - 1}}$ is
 (A) 0 (B) $\frac{1}{2}$
 (C) 1 (D) $\frac{1}{\sqrt{2}}$
4. Let $g(x) = \max|y^2 - xy|$, $y \in [0, 1]$, then the absolute minimum value of $g(x)$ is
 (A) 1 (B) 0
 (C) $3 - 2\sqrt{2}$ (D) $2\sqrt{2} - 2$
5. $\lim_{n \rightarrow \infty} \left(\binom{n}{0} \binom{n}{1} \binom{n}{2} \dots \binom{n}{n} \right)^{\frac{1}{n(n+1)}}$ is
 (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$
 (C) $\frac{1}{\sqrt{e}}$ (D) $\sqrt{e}$

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
6. Let Z be a variable complex number satisfying $\frac{2z - \bar{z}}{3} = a - \frac{1}{3} + ib$ where a, b are real parameters such that $a^2 - b^2 = 1$. If locus of z is a curve 'C', then
- (A) director circle of the curve 'C' is $(x + 1)^2 + y^2 = 10$
- (B) area of the triangle formed by the straight line $x - 3y + 1 = 0$ and tangent to the curve 'C' at one of its vertex and the x-axis is $\frac{3}{2}$ sq. units
- (C) Distance between directrices of the curve 'C' is $\frac{18}{\sqrt{10}}$
- (D) Equation of the circle passing through the points (2,1) and both foci of the curve 'C' is $x^2 + y^2 + 2x - 9 = 0$

7. If M_1 and M_2 are the feet of perpendiculars from foci F_1 and F_2 of the ellipse $\frac{x^2}{64} + \frac{y^2}{25} = 1$ on the tangent at any point P of the ellipse respectively. If $M_1F_1 = p$ and $M_2F_2 = q$ then
- (A) $\min(p^2 + q^2) = 50$
 (B) $\min(p^2 + q^2) = 25$
 (C) $\min(p + q) = 10$
 (D) $\min(p + q) = 50$
8. Let $AB : y = x$ be a chord to the parabola $y^2 + 4x = 4$ and P be any variable point on the ellipse $x^2 + 9(y-4)^2 - 9 = 0$. If Δ_1 and Δ_2 be the minimum and maximum area of triangle ABP respectively then
- (A) $\Delta_2 = 2\sqrt{2}(4 + \sqrt{10})$
 (B) $\Delta_2 - \Delta_1 = 8\sqrt{5}$
 (C) $\frac{\Delta_2}{\Delta_1} = \frac{13 + 4\sqrt{10}}{3}$
 (D) $\Delta_2 - \Delta_1 = 5\sqrt{5}$
9. $f(x) = \lim_{n \rightarrow \infty} \sum_{r=1}^n 3^{r-1} \sin^3 \frac{x}{3^r}$, then
- (A) $f\left(\frac{\pi}{2}\right) = \frac{\pi}{8} - \frac{1}{4}$
 (B) $f\left(\frac{\pi}{2}\right) = \frac{\pi}{8}$
 (C) $f\left(\frac{3\pi}{2}\right) = \frac{3\pi}{8}$
 (D) $f\left(\frac{3\pi}{2}\right) = \frac{3\pi}{8} + \frac{1}{4}$
10. Let $f(x, y, z) = xyz$ and $A = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1, x + y + z = 1\}$. Then over set A
- (A) global maximum of $f(x, y, z)$ is 0
 (B) global maximum of $f(x, y, z)$ is $-\frac{4}{27}$
 (C) global maximum of x is 1
 (D) global minimum of z is $-\frac{1}{3}$

11. Let $m, n \in \mathbb{N}$ such that $n > m > 2$. Consider a cubic equation $mx^3 + nx^2 - 1 = 0$ with α, β, γ as its roots. Which of the following statements are true ?
- (A) All three roots α, β, γ are real.
- (B) Two roots are negative and one is positive
- (C) If $\alpha < \beta < \gamma$ then $|\gamma| < |\beta| < |\alpha|$
- (D) If $\alpha < \beta < \gamma$ then $|\beta| < |\gamma| < |\alpha|$
12. Let f be a differentiable function on $[0, 1]$ such that $f(0) = 0$ and $f(1) = 1$. Then
- (A) There exist $x_1, x_2 \in (0, 1)$ and $x_1 \neq x_2$ such that $f'(x_1) + f'(x_2) = 2$
- (B) There exist $x_1, x_2 \in (0, 1)$ and $x_1 \neq x_2$ such that $\frac{1}{f'(x_1)} + \frac{1}{f'(x_2)} = 2$
- (C) Corresponding to arbitrary positive real numbers k_1 and k_2 , there exist $x_1, x_2 \in (0, 1)$ and $x_1 \neq x_2$ such that $\frac{k_1}{f'(x_1)} + \frac{k_2}{f'(x_2)} = k_1 + k_2$.
- (D) There exists n distinct real numbers $x_1, x_2, \dots, x_n \in (0, 1)$ such that $f'(x_1) + f'(x_2) + \dots + f'(x_n) = n$
13. Consider two curves namely $C_1 : y = a^x$ and $C_2 : x = a^y$; $a > 0$ then
- (A) Number of times C_1 meets with C_2 is 3 for $a \in (0, e^{-e})$
- (B) Number of times C_1 meets with C_2 is 1 for $a \in [e^{-e}, 1] \cup \{e^{e^{-1}}\}$
- (C) Number of times C_1 meets with C_2 is 2 for $a \in (1, e^{e^{-1}})$
- (D) Number of times C_1 meets with C_2 is 0 for $a \in (e^{e^{-1}}, \infty)$

Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct answer is chosen.

Zero Marks : 0 In all other cases.

1. For $a, b \in \mathbb{R}^+$, $\int_a^b \frac{e^{\frac{x}{a}} - e^{\frac{x}{b}}}{x} dx$ is
2. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B = \begin{bmatrix} p \\ q \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ such that $AB = B$ and $\text{tr.}(A) = 50$ and $ad - bc$ is k then $\lfloor \sqrt{k} \rfloor$ is
3. Let $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$. If sum of all elements of A^{10} is 'k' then $\left\lfloor \frac{k}{2560} \right\rfloor$ is
4. Number of terms which are odd in the sequence $a_0, a_1, a_2, \dots, a_{2017}$ where $a_n = \binom{2017}{n}$ is $\frac{k}{2}$. Then $\lfloor \sqrt{k} \rfloor$ is
5. Number of non constant differentiable function $f: \mathbb{R} \rightarrow \mathbb{R}^+ \cup \{0\}$, such that $f(f(f(x))) = f(x), \forall x \in \mathbb{R}$ is

Space for Rough Work

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